

BCA-SEMESTER 2ND SYLLABUS

1) Solution of system of linear equations

- a) Direct Method
 - i) Gauss Elimination method
 - ii) Gauss Jordan Method
 - iii) LU Decomposition Method
- b) Indirect (Iterative) Method
 - i) Gauss Jacobi Method
 - ii) Gauss Seidel Method

2) Solution of Algebraic and Transcendental equation

- a) Newton Rapsons Method
- b) Iterative Method
- c) Regula Falsi Method
- d) Bisection (bolzano) Method

3) Interpolation Methods

- a) Lagrange Interpolation method
- b) Newton Forward Interpolation
- c) Newton Backward Interpolation
- d) Central Intermolation

4) Numerical Differentiation

5) Numerical Integration

- a) Trapeqoidal Rule
- b) Simpson's 1/3 Rule
- c) Simpson's 3/8 Rule

6) Solution of ordinary differential equation

- d) Tatlor Series Method
- e) Euler's Method
- f) Modified Euler's Method
- g) Runge – Kutta Method (2nd order)
- h) Runge – Kutta Method (4nd order)

7) Error Analysis

- a) Absalute Error
- b) Relative Error
- c) Percentage Error
- d) Significant Error

NUMERICAL INTEGRATION

- Numerical Integration is the process to calculate the value of definite integration

$$\int_a^b f(x) dx$$

- Some method for solving numerical integration.

i) Trapezoidal Rule

ii) Simpson's 1/3 Rule

iii) Simpson's 3/8 Rule

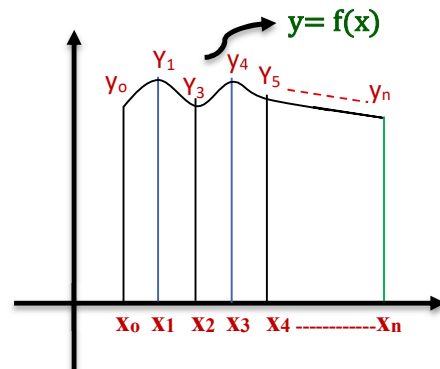
TRAPEZOIDAL RULE

(THEORY)

- ❖ Explain trapezoidal rule for solving definite integration (Numerical Integration)

➤ Now,

we divide the graph of $f(x)$ into (n) number of small trapezium with height of is trapezium small



$$h = \frac{(b-a)}{n} \text{ where,}$$

n = no of sub interval

$$\int_a^b f(x) dx = \text{Area}(\text{trap}(x_0 y_0 y_1 x_1)) + \text{Area}(\text{trap}(x_1 y_1 y_2 x_2)) + \dots + \text{Area}(\text{trap}(x_{n-1} y_{n-1} y_n x_n))$$

$$\Rightarrow \frac{h}{2}(y_0 + y_1) + \frac{h}{2}(y_1 + y_2) + \frac{h}{2}(y_2 + y_3) + \dots + \frac{h}{2}(y_{n-1} + y_n)$$

$$\Rightarrow \frac{h}{2}[(y_0 + y_n) + 2(y_1 + y_2 + y_3 \dots y_{n-1})]$$

(PRACTICAL - 1)

Q1. Find the definite integration by using trapezoidal rule :-

$$\int_0^1 \frac{dx}{1+x^2}$$

Soln

Given that,

$$F(x) = \frac{1}{1+x^2}$$

$$a = 0, \quad b = 1, \quad n = 4$$

let,

$$h = \frac{b-a}{n} = \frac{1-0}{4} = \frac{1}{4}$$

$$h = \frac{1}{4}$$

x	0	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{3}{4}$	1
F(x)	1	$\frac{16}{4}$	$\frac{4}{5}$	$\frac{16}{25}$	$\frac{1}{2}$
	y ₀	y ₁	y ₂	y ₃	y ₄

By - Trapezoidal rule :-

$$\triangleright \int_0^1 \frac{dx}{1+x^2} = \frac{h}{2} [(y_0 + y_4) + 2(y_1 + y_2 + y_3)]$$

$$= \frac{1}{4 \times 2} [(1 + \frac{1}{2}) + 2(\frac{16}{17} + \frac{4}{5} + \frac{16}{25})]$$

$$= \frac{1}{8} [(\frac{3}{2} + \frac{32}{17} + \frac{8}{5} + \frac{32}{25})]$$

$$= \frac{1}{8}(6.2623)$$

$$= 0.7826$$

Answer

(PRACTICAL - 2)

Q2. Find the definite integration by using trapezoidal rule :-

$$\int_0^{\pi/4} \sin 4x \, dx$$

Soln

Given that,

$$F(x) = \sin 4x \, dx$$

$$a = 0, \quad b = \pi/4, \quad n = 4$$

let,

$$h = \frac{b-a}{n} = \frac{\pi/4-0}{4} = 0.196$$

$$h = 0.196$$

x	0	0.196	0.392	0.588	0.785
F(x)	0	0.706	1	0.710	0
	y_0	y_1	y_2	y_3	y_4

By - Trapezoidal rule :-

$$\begin{aligned} \Rightarrow \int_0^1 \frac{dx}{1+x^2} &= \frac{h}{2} [(y_0 + y_4) + 2(y_1 + y_2 + y_3)] \\ &= \frac{0.196}{2} [(0 + 0) + 2(0.706 + 1 + 0.710)] \\ &= 0.098 [(0 + 1.412 + 2 + 1.42)] \\ &= 0.098(4.832) \\ &= 0.473536 \end{aligned}$$

Answer

(PRACTICAL - 3)

Q3. Find the definite integration by using trapezoidal rule :-

$$\int_0^6 \frac{dx}{1+x^2}$$

Soln

Given that,

$$F(x) = \frac{1}{1+x^2}$$

$$a = 0, \quad b = 6, \quad n = 4$$

let,

$$h = \frac{b-a}{n} = \frac{6-0}{4} = 1.5$$

$$h = 1.5$$

x	0	1.5	3	4.5	6
F(x)	1	0.3077	0.1	0.0471	0.027
	y_0	y_1	y_2	y_3	y_4

By - Trapezoidal rule :-

$$\begin{aligned} \triangleright \int_0^6 \frac{dx}{1+x^2} &= \frac{h}{2} [(y_0 + y_4) + 2(y_1 + y_2 + y_3)] \\ &= \frac{1.5}{2} [(1 + 0.027) + 2(0.3077 + 0.1 + 0.0471)] \\ &= 0.75 [(1.027 + 0.9096)] \\ &= 0.75(1.9366) \\ &= 1.4524 \end{aligned}$$

Answer

(EXERCISE - 1)

(Objectives)

❖ Trapezoidal Rule – Multiple Choice Questions :-

1. What is the primary use of the Trapezoidal Rule?
 - A) Differentiation
 - B) Integration
 - C) Factorization
 - D) Interpolation

2. The Trapezoidal Rule approximates the area under a curve by dividing it into:
 - A) Triangles
 - B) Rectangles
 - C) Trapezoids
 - D) Circles

3. In the Composite Trapezoidal Rule, the step size h is given by:
 - A) $(a + b)/n$
 - B) $(b - a)/n$
 - C) ab/n
 - D) $(a - b)/n$

4. Which of the following is the formula for the Single Trapezoidal Rule?
 - A) $(b - a)[f(a) + f(b)]$
 - B) $\frac{1}{2}(b - a)[f(a) + f(b)]$
 - C) $(b + a)[f(a) - f(b)]$
 - D) $\frac{1}{2}(a + b)[f(a) + f(b)]$

5. In the Composite Trapezoidal Rule, the first and last ordinates are counted:
 - A) Twice
 - B) Three Times
 - C) Once
 - D) Four Times

(EXERCISE - 2)

(Subjectives)

❖ Trapezoidal Rule – Numerical Questions :-

Q1. Find the definite integration by using Trapezoidal Rule:

$$\int_0^1 \frac{dx}{1+x^2}$$

Q2. Find the definite integration by using Trapezoidal Rule:

$$\int_0^2 (x^2 + 1) dx$$

Q3. Find the definite integration by using Trapezoidal Rule:

$$\int_0^4 x^2 dx$$

Q4. Find the definite integration by using Trapezoidal Rule:

$$\int_1^3 (x^3 + 2x) dx$$

Q5. Find the definite integration by using Trapezoidal Rule:

$$\int_0^1 e^x dx$$

Q6. Find the definite integration by using Trapezoidal Rule:

$$\int_0^\pi \sin x dx$$

Q7. Find the definite integration by using Trapezoidal Rule:

$$\int_1^2 \frac{dx}{x}$$

Note: Use $n = 5$ in all numerical problems.

SIMPSON'S $\frac{1}{3}$ RULE

(THEORY)

❖ Explain Simpson's $\frac{1}{3}$ rule for solving definite integration (Numerical Integration)

➤ Let,

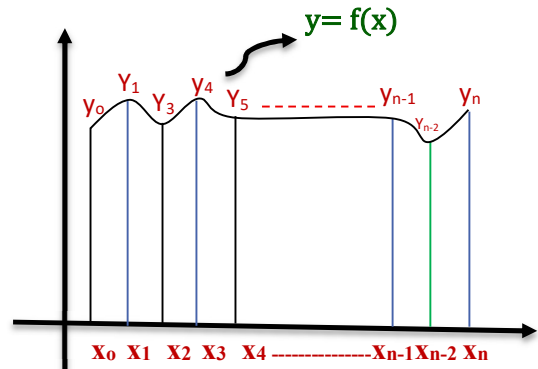
$f(x)$ be the real value continuous function defined on closed interval $[a, b]$. We have to find

$$\int_a^b f(x) dx$$

We divide the graph of $f(x)$ into

n (number) of partitions of parabola, where

n is the number corresponding to each pair of sub intervals.



➤ $\int_a^b f(x) dx = \text{Area}(\text{Para}(x_0, y_0, y_1, y_2, x_2)) + \text{Area}(\text{Para}(x_2, y_2, y_3, y_4, x_4)) + \text{Area}(\text{Para}(x_4, y_4, y_5, y_6, x_6)) + \dots + \text{Area}(\text{Para}(x_{n-2}, y_{n-2}, y_{n-1}, y_n, x_n))$

$$= \frac{h}{3}(y_0 + 4y_1 + y_2) + \frac{h}{3}(y_2 + 4y_3 + y_4) + \frac{h}{3}(y_4 + 4y_5 + y_6) + \dots + \frac{h}{3}(y_{n-2} + 4y_{n-1} + y_n)$$

$$= \frac{h}{3}[(y_0 + y_n) + 4(y_1 + y_3 + y_5 + \dots + y_{n-1}) + 2(y_2 + y_4 + y_6 + y_8 + \dots + y_{n-2})]$$

Hence,

$$\int_a^b f(x) dx = \frac{h}{3}[(y_0 + y_n) + 4(y_1 + y_3 + y_5 + \dots + y_{n-1}) + 2(y_2 + y_4 + y_6 + y_8 + \dots + y_{n-2})]$$

where,

$$h = \frac{b - a}{n}$$

(PRACTICAL - 1)

Q1. Evaluate the definite integration by using Simpson's $\frac{1}{3}$ rule :-

$$\int_0^1 \frac{dx}{1+x^2}$$

Soln

Given that,

$$F(x) = \frac{1}{1+x^2}$$

$$a = 0, \quad b = 1, \quad n = 6$$

let,

$$h = \frac{b-a}{n} = \frac{1-0}{6} = \frac{1}{6}$$

$$h = \frac{1}{6}$$

x	0	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{2}$	$\frac{2}{3}$	$\frac{5}{6}$	1
F(x)	1	$\frac{36}{37}$	$\frac{9}{10}$	$\frac{4}{5}$	$\frac{9}{13}$	$\frac{36}{61}$	$\frac{1}{2}$
	y_0	y_1	y_2	y_3	y_4	y_5	y_6

By - Simpson's $\frac{1}{3}$ rule :-

$$\triangleright \int_0^1 \frac{dx}{1+x^2} = \frac{h}{3} [(y_0 + y_6) + 4(y_1 + y_3 + y_5) + 2(y_2 + y_4)]$$

$$= \frac{1}{3 \times 6} [(1 + \frac{1}{2}) + 4(\frac{36}{37} + \frac{4}{5} + \frac{36}{61}) + 2(\frac{9}{10} + \frac{9}{13})]$$

$$= \frac{1}{18} [(\frac{3}{2} + \frac{144}{37} + \frac{16}{5} + \frac{144}{61} + \frac{18}{10} + \frac{18}{13})]$$

$$= \frac{1}{18} (14.137163)$$

$$= 0.785397945234$$

Answer

(PRACTICAL - 2)

Q2. Evaluate the definite integration by using Simpson's $\frac{1}{3}$ rule :-

$$\int_0^6 \frac{dx}{1+x^2}$$

Soln

Given that,

$$F(x) = \frac{1}{1+x^2}$$

$$a = 0, \quad b = 6, \quad n = 6$$

let,

$$h = \frac{b-a}{n} = \frac{6-0}{6} = 1$$

$$h = 1$$

x	0	2	2	3	4	5	6
F(x)	1	0.5	0.2	0.1	0.0588	0.0385	0.027
	y_0	y_1	y_2	y_3	y_4	y_5	y_6

By - Simpson's $\frac{1}{3}$ rule :-

$$\text{➤ } \int_0^6 \frac{dx}{1+x^2} = \frac{h}{3} [(y_0 + y_6) + 4(y_1 + y_3 + y_5) + 2(y_2 + y_4)]$$

$$= \frac{1}{3} [(1 + 0.027) + 4(0.5 + 0.1 + 0.0385) + 2(0.2 + 0.0588)]$$

$$= \frac{1}{3} (4.0986)$$

$$= 1.3662$$

Answer

(PRACTICAL - 3)

Q3. Evaluate the definite integration by using Simpson's $\frac{1}{3}$ rule :-

$$\int_0^1 \frac{dx}{1+x^2}$$

Soln

Given that,

$$F(x) = \frac{1}{1+x^2}$$

$$a = 0, \quad b = 1, \quad n = 6$$

let,

$$h = \frac{b-a}{n} = \frac{1-0}{6} = \frac{1}{6}$$

$$h = \frac{1}{6}$$

x	0	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{2}$	$\frac{2}{3}$	$\frac{5}{6}$	1
F(x)	1	$\frac{36}{37}$	$\frac{9}{10}$	$\frac{4}{5}$	$\frac{9}{13}$	$\frac{36}{61}$	$\frac{1}{2}$
	y_0	y_1	y_2	y_3	y_4	y_5	y_6

By - Simpson's $\frac{1}{3}$ rule :-

$$\triangleright \int_0^1 \frac{dx}{1+x^2} = \frac{h}{3} [(y_0 + y_6) + 4(y_1 + y_3 + y_5) + 2(y_2 + y_4)]$$

$$= \frac{1}{3 \times 6} [(1 + \frac{1}{2}) + 4(\frac{36}{37} + \frac{4}{5} + \frac{36}{61}) + 2(\frac{9}{10} + \frac{9}{13})]$$

$$= \frac{1}{18} [(\frac{3}{2} + \frac{144}{37} + \frac{16}{5} + \frac{144}{61} + \frac{18}{10} + \frac{18}{13})]$$

$$= \frac{1}{18} (14.137163)$$

$$= 0.785397945234$$

Answer

(EXERCISE - 1)

(Objectives)

❖ Simpson's 1/3 Rule – Multiple Choice Questions :-

Q1. In Simpson's 1/3 Rule, the number of sub-intervals n must be:

- A) Odd number
- B) Prime number
- C) Even number
- D) Multiple of 3

Q2. The factor outside the bracket in Simpson's 1/3 Rule is:

- A) $\frac{h}{2}$
- B) $\frac{h}{3}$
- C) $\frac{3h}{8}$
- D) $\frac{h}{4}$

Q3. Simpson's 1/3 Rule is used for:

- A) Numerical Differentiation
- B) Numerical Integration
- C) Matrix Multiplication
- D) Interpolation Only

Q4. In Simpson's 1/3 Rule, the coefficient of the odd ordinates is:

- A) 1
- B) 2
- C) 3
- D) 4

Q5. The width of each sub-interval in Simpson's 1/3 Rule is given by:

- A) $h = \frac{n}{b-a}$
- B) $h = \frac{a+b}{n}$
- C) $h = \frac{b-a}{n}$
- D) $h = \frac{ab}{n}$

(EXERCISE - 2)

(Subjectives)

❖ **Simpson's 1/3 Rule – Numerical Questions :-**

Q1. Explain Simpson's 1/3 Rule for Numerical Integration and derive its formula.

Q2. Find the definite integration by using Simpson's 1/3 Rule:

$$\int_0^2 \frac{1}{1+x} dx$$

Q3. Find the definite integration by using Simpson's 1/3 Rule:

$$\int_0^2 x^2 dx$$

Q4. Find the definite integration by using Simpson's 1/3 Rule:

$$\int_0^2 e^x dx$$

Q5. Find the definite integration by using Simpson's 1/3 Rule:

$$\int_0^2 \sin x dx$$

Q6. Find the definite integration by using Simpson's 1/3 Rule:

$$\int_0^2 \sqrt{1+x} dx$$

Q7. Find the definite integration by using Simpson's 1/3 Rule:

$$\int_0^2 \frac{1}{1+x^2} dx$$

Note: Use $n = 2$ in all numerical problems.

SIMPSON'S $\frac{3}{8}$ RULE

(THEORY)

❖ Explain Simpson's $\frac{3}{8}$ rule for solving definite integration (Numerical Integration)

➤ Let,

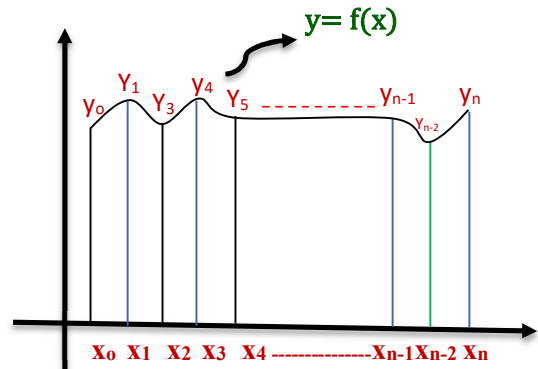
$f(x)$ be the real value continuous function defined on closed interval $[a, b]$. We have to find

$$\int_a^b f(x) dx$$

We divide the graph of $f(x)$ into

n (number) of partitions of parabola, where

n is the number corresponding to each pair of sub intervals.



➤ $\int_a^b f(x) dx = \text{Area}(\text{Para}(x_0, y_0, y_1, y_2, y_3, x_3)) + \text{Area}(\text{Para}(x_3, y_3, y_4, y_5,$

$y_6, x_6)) + \text{Area}(\text{Para}(x_6, y_6, y_7, y_8, y_9, x_9)) + \dots + \text{Area}$

$(\text{Para}(x_{n-2}, y_{n-2}, y_{n-1}, y_n, x_n))$

$$= \frac{3h}{8} (y_0 + 3y_1 + 3y_2 + y_3) + \frac{3h}{8} (y_3 + 3y_4 + 3y_5 + y_6) + \frac{3h}{8} (y_6 + 3y_7 + 3y_8 + y_9) + \dots + \frac{3h}{8} (y_{n-3} + 3y_{n-2} + 3y_{n-1} + y_n)$$

Hence,

$$\int_a^b f(x) dx = \frac{3h}{8} [(y_0 + y_n) + 3(y_1 + y_4 + y_7 + \dots + y_{n-1}) + 3(y_2 + y_5 + y_8 + \dots + y_{n-2}) + 2(y_3 + y_6 + y_9 + \dots + y_{n-3})]$$

where,

$$h = \frac{b-a}{n}$$

(PRACTICAL - 1)

Q1. Evaluate the definite integration by using Simpson's $\frac{3}{8}$ rule :-

$$\int_0^1 \frac{dx}{1+x^2}$$

Soln

Given that,

$$F(x) = \frac{1}{1+x^2}$$

$$a = 0, \quad b = 1, \quad n = 6$$

let,

$$h = \frac{b-a}{n} = \frac{1-0}{6} = \frac{1}{6}$$

$$h = \frac{1}{6}$$

x	0	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{2}$	$\frac{2}{3}$	$\frac{5}{6}$	1
F(x)	1	$\frac{36}{37}$	$\frac{9}{10}$	$\frac{4}{5}$	$\frac{9}{13}$	$\frac{36}{61}$	$\frac{1}{2}$
	y_0	y_1	y_2	y_3	y_4	y_5	y_6

By - Simpson's $\frac{1}{3}$ rule :-

$$\triangleright \int_0^1 \frac{dx}{1+x^2} = \frac{3h}{8} [(y_0 + y_6) + 3(y_1 + y_4) + 3(y_2 + y_5) + 2(y_3)]$$

$$= \frac{3 \cdot 1/6}{8} [(1 + \frac{1}{2}) + 3(\frac{36}{37} + \frac{9}{13}) + 3(\frac{9}{10} + \frac{36}{61}) + 2(\frac{4}{5})]$$

$$= \frac{1}{16} [(\frac{3}{2} + 3(1.66) + 3(1.49) + 2(\frac{4}{5}))]$$

$$= \frac{1}{16} (\frac{3}{2} + \frac{8}{5} + 4.98 + 4.47)$$

$$= 0.7843$$

Answer

(PRACTICAL - 2)

Q2. Evaluate the definite integration by using Simpson's $\frac{3}{8}$ rule :-

$$\int_0^1 \frac{dx}{1+x^2}$$

Soln

Given that,

$$F(x) = \frac{1}{1+x^2}$$

$$a = 0, \quad b = 1, \quad n = 6$$

let,

$$h = \frac{b-a}{n} = \frac{1-0}{6} = \frac{1}{6}$$

$$h = \frac{1}{6}$$

x	0	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{2}$	$\frac{2}{3}$	$\frac{5}{6}$	1
F(x)	1	$\frac{36}{37}$	$\frac{9}{10}$	$\frac{4}{5}$	$\frac{9}{13}$	$\frac{36}{61}$	$\frac{1}{2}$
	y_0	y_1	y_2	y_3	y_4	y_5	y_6

By - Simpson's $\frac{1}{3}$ rule :-

$$\triangleright \int_0^1 \frac{dx}{1+x^2} = \frac{3h}{8} [(y_0 + y_6) + 3(y_1 + y_4) + 3(y_2 + y_5) + 2(y_3)]$$

$$= \frac{3 \cdot 1/6}{8} [(1 + \frac{1}{2}) + 3(\frac{36}{37} + \frac{9}{13}) + 3(\frac{9}{10} + \frac{36}{61}) + 2(\frac{4}{5})]$$

$$= \frac{1}{16} [(\frac{3}{2} + 3(1.66) + 3(1.49) + 2(\frac{4}{5}))]$$

$$= \frac{1}{16} (\frac{3}{2} + \frac{8}{5} + 4.98 + 4.47)$$

$$= 0.7843$$

Answer

(PRACTICAL - 3)

Q1. Evaluate the definite integration by using Simpson's $\frac{3}{8}$ rule :-

$$\int_0^1 \frac{dx}{1+x^2}$$

Soln

Given that,

$$F(x) = \frac{1}{1+x^2}$$

$$a = 0, \quad b = 1, \quad n = 6$$

let,

$$h = \frac{b-a}{n} = \frac{1-0}{6} = \frac{1}{6}$$

$$h = \frac{1}{6}$$

x	0	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{2}$	$\frac{2}{3}$	$\frac{5}{6}$	1
F(x)	1	$\frac{36}{37}$	$\frac{9}{10}$	$\frac{4}{5}$	$\frac{9}{13}$	$\frac{36}{61}$	$\frac{1}{2}$
	y_0	y_1	y_2	y_3	y_4	y_5	y_6

By - Simpson's $\frac{1}{3}$ rule :-

$$\triangleright \int_0^1 \frac{dx}{1+x^2} = \frac{3h}{8} [(y_0 + y_6) + 3(y_1 + y_4) + 3(y_2 + y_5) + 2(y_3)]$$

$$= \frac{3 \cdot 1/6}{8} [(1 + \frac{1}{2}) + 3(\frac{36}{37} + \frac{9}{13}) + 3(\frac{9}{10} + \frac{36}{61}) + 2(\frac{4}{5})]$$

$$= \frac{1}{16} [(\frac{3}{2} + 3(1.66) + 3(1.49) + 2(\frac{4}{5}))]$$

$$= \frac{1}{16} (\frac{3}{2} + \frac{8}{5} + 4.98 + 4.47)$$

$$= 0.7843$$

Answer

(EXERCISE - 1)

(Objectives)

❖ Simpson's 3/8 Rule – Multiple Choice Questions :-

Q1. In Simpson's 3/8 Rule, the coefficient of y_1 is:

- A) 1
- B) 2
- C) 3
- D) 4

Q2. The factor outside the bracket in Simpson's 3/8 Rule is:

- A) $\frac{h}{3}$
- B) $\frac{3h}{8}$
- C) $\frac{h}{2}$
- D) $\frac{8h}{3}$

Q3. Simpson's 3/8 Rule is a method of:

- A) Numerical Differentiation
- B) Numerical Integration
- C) Matrix Analysis
- D) Interpolation Only

Q4. For Simpson's 3/8 Rule, the number of sub-intervals n should be:

- A) Even
- B) Odd
- C) Multiple of 2
- D) Multiple of 3

Q5. The general formula of Simpson's 3/8 Rule begins with:

- A) $\frac{h}{2}$
- B) $\frac{h}{3}$
- C) $\frac{3h}{8}$
- D) $\frac{8h}{3}$

(EXERCISE - 2)

(Subjectives)

❖ Simpson's 3/8 Rule – Numerical Questions :-

Q1. Explain Simpson's 3/8 Rule for Numerical Integration and derive its formula.

Q2. Find the definite integration by using Simpson's 3/8 Rule:

$$\int_0^3 \frac{1}{1+x} dx$$

Q3. Find the definite integration by using Simpson's 3/8 Rule:

$$\int_0^3 x^2 dx$$

Q4. Find the definite integration by using Simpson's 3/8 Rule:

$$\int_0^3 e^x dx$$

Q5. Find the definite integration by using Simpson's 3/8 Rule:

$$\int_0^3 \sin x dx$$

Q6. Find the definite integration by using Simpson's 3/8 Rule:

$$\int_0^3 \sqrt{1+x} dx$$

Q7. Find the definite integration by using Simpson's 3/8 Rule:

$$\int_0^3 \frac{1}{1+x^2} dx$$

Note: Use $n = 3$ in all numerical problems.

